ModelIt!

Teacher's Guide

Renewable Energy Grid Optimization: Powering the Future

Modeling the Complex Challenge of a 100 Percent Clean Energy Grid

Quick Reference

Grade Level	Grade 09L2
NGSS Standard	HS-PS3-3, HS-ETS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS3-3, HS-ETS1-3

Design, build, and refine a device that works within given constraints to convert one form of energy into another. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students optimize multi-variable energy systems, balancing competing objectives using computational models and scenario analysis.
Disciplinary Core Idea	**PS3.D Energy in Chemical Processes and Everyday Life** — Energy transformation from solar radiation and wind kinetic energy to electrical energy involves efficiency losses and intermittency challenges.
Crosscutting Concept	**Energy and Matter** — Students model energy flows through a grid system, tracking generation, storage, transmission losses, and consumption.

CAST Testing Alignment

MULTIPLE CHOICE:

Which factor MOST limits how much renewable energy a grid can use: (a) panel/turbine cost, (b) storage cost and capacity, (c) transmission distance, (d) public opinion?

CONSTRUCTED RESPONSE:

Using your model, explain why a grid with 80 percent renewables might cost twice as much as one with 50 percent renewables, even though the additional solar panels are cheap.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The clean energy transition is the defining engineering challenge of our generation. While renewable energy costs have plummeted, the fundamental challenge of intermittency remains: the sun does not always shine, the wind does not always blow, and society demands electricity 24/7.

The Intermittency Challenge

Solar produces maximum power at noon but zero at night. Wind varies hourly and seasonally. Grid operators must match supply to demand in real time. Even a few minutes of mismatch causes blackouts. Fossil fuel plants can be turned on and off as needed -- renewables cannot.

The Storage Gap

Current battery technology can bridge hours of intermittency but not days or weeks. A week-long calm, cloudy period would exhaust any economically feasible battery system. Alternative storage like pumped hydro and hydrogen is promising but limited by geography and cost.

The Last 20 Percent Problem

Getting to 50-60 percent renewables is relatively straightforward. Getting to 80 percent requires massive storage. Getting to 100 percent requires solving multi-day and seasonal intermittency gaps that current technology cannot economically address. Each additional percent of renewables costs more than the last.

Grid Physics

Electricity must be consumed the instant it is generated. The grid operates at exactly 60 Hz and must be balanced continuously. Too much supply raises frequency; too little drops it. Either extreme damages equipment and causes cascading failures. This physical constraint makes grid management fundamentally different from other optimization problems.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify solar output, wind output, battery storage, grid demand, fossil backup, transmission efficiency, cost per kWh, and carbon emissions as the 8 grid components.

E **Establish Relationships**
Students map supply-demand dynamics: renewables feed demand directly, excess charges batteries, batteries discharge when needed, fossil backup fills remaining gaps.

V **Visualize & Model**
Students build a model optimizing the trade-offs between cost, emissions, and reliability across different energy mixes.

E **Evaluate Outcomes**
Students run sunny, worst-case, and optimization scenarios to find the best achievable energy mix.

R **Revise & Extend**
Students add hydrogen storage, nuclear baseload, or demand response to explore advanced grid strategies.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Why can't we just flip a switch to clean energy? The answer is more complicated than you think."*

Watch For

Show the intermittency problem visually: solar output over 24 hours vs. demand.

Slide 2: Learning Objectives

3 min

Say: *"Today we optimize the most complex engineering problem of our generation."*

Watch For

Pre-teach intermittency and grid balancing.

Slide 3: Big Question

3 min

Say: *"Solar panels are cheap. Wind turbines work great. So why are we still burning fossil fuels?"*

Watch For

Quick-write: What happens to electricity supply at night?

Slide 4: LEVER Framework

2 min

Say: *"Eight components. Three competing objectives. This is real engineering."*

Watch For

Preview the optimization challenge: minimize cost, minimize emissions, maximize reliability.

Slide 5: Activity 1: Components

8 min

Say: *"Which of these can we control? Which are set by physics?"*

Watch For

Sort components. Discuss why weather determines solar and wind output.

Slide 6: Activity 2: Connect

10 min

Say: *"Energy flows from generation through storage to demand. Map the pathways."*

Watch For

Map energy flows. Identify where gaps occur and how each component fills them.

Slide 7: Activity 3: Simulate

10 min

Say: *"Find the mix that keeps the lights on AND saves the planet. What trade-offs do you make?"*

Watch For

Students optimize energy mix across competing objectives.

Slide 8: Discoveries

5 min

Say: *"The last 20 percent is the hardest. Your model shows you why."*

Watch For

Discuss the exponential difficulty curve and what technology changes would help.

Slide 9: STEM Challenge

12 min

Say: *"5 million people need 90 percent clean energy by 2035. Design the grid."*

Watch For

Groups design complete energy portfolios with cost and emissions analysis.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Grid Demand, Fossil Fuel Backup
- **INTERNAL:** Solar Output, Wind Output, Battery Storage, Transmission Efficiency, Cost per kWh, Carbon Emissions

Relationships

- **Solar Output to Battery Storage:** POSITIVE (+) — Excess solar output beyond demand charges batteries.
- **Wind Output to Battery Storage:** POSITIVE (+) — Excess wind output charges batteries.
- **Grid Demand to Battery Storage:** NEGATIVE (-) — Higher demand depletes battery reserves faster.
- **Battery Storage to Fossil Fuel Backup:** NEGATIVE (-) — More available storage reduces the need for fossil backup.
- **Fossil Fuel Backup to Carbon Emissions:** POSITIVE (+) — More fossil fuel use directly increases carbon emissions.
- **Fossil Fuel Backup to Cost per kWh:** POSITIVE (+) — Fossil fuel operation adds fuel costs to the electricity price.
- **Battery Storage to Cost per kWh:** POSITIVE (+) — Storage infrastructure is expensive, raising overall cost.
- **Transmission Efficiency to Cost per kWh:** NEGATIVE (-) — Higher efficiency reduces waste and lowers per-unit cost.

Key Discoveries

- The last 10-20 percent of fossil fuel replacement is exponentially harder than the first 80 percent due to intermittency gaps
- Battery storage is essential but currently too expensive to cover multi-day renewable droughts without fossil backup
- Overbuilding renewable capacity to handle worst-case scenarios dramatically increases cost but reduces emissions
- Geographic diversity helps -- connecting solar-rich and wind-rich regions reduces the probability of system-wide intermittency

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Renewables are too expensive"	Solar and wind are now the CHEAPEST sources of new electricity generation. The cost challenge is in STORAGE and grid integration, not generation. The total system cost including storage is higher than generation alone, but still competitive with new fossil fuel plants in most regions.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"We can go 100 percent renewable immediately"	The intermittency problem makes 100 percent renewables extremely challenging with current technology. Multi-day and seasonal gaps require storage capacity that does not yet exist at affordable scale. A rapid transition without adequate storage would cause blackouts.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Nuclear is too dangerous for clean energy"	Modern nuclear power has the lowest death rate per unit of energy of any source, including solar and wind. It provides reliable, zero-carbon baseload power that complements intermittent renewables. The main challenges are cost, construction time, and waste storage -- not safety.